

Part II

Functional Analysis and Operator Theory

Chapter 5

Functional Analysis

Functional analysis extends the finite-dimensional linear algebra of Chapter 2 to infinite-dimensional spaces. In quantum information, these tools are essential for continuous variable systems (harmonic oscillators, quantum optics, bosonic systems) and for the structural theory of quantum channels. This chapter introduces the core functional analytic framework: normed and Banach spaces, Hilbert spaces, bounded operators, dual spaces, the Riesz representation theorem, weak topologies, and L^p spaces.

The reader is assumed to be familiar with the linear algebra reviewed in Chapter 2 and the analysis and measure theory reviewed in Chapters 1 and 3. The operator theory developed in this chapter will be extended in Chapter 7, where we study adjoints, spectra, compact operators, trace-class operators, and functional calculus.

Throughout this chapter, all vector spaces are over $\mathbb{C}$ unless stated otherwise.

5.1 Metric and Topological Spaces

Metric and topological spaces provide the language of convergence, continuity, and approximation. The finite-dimensional case was covered in Chapter 1; here we briefly recall the definitions needed for infinite-dimensional spaces.

Definition 5.1.1. A *metric space* is a set X together with a function $d : X \times X \rightarrow [0, \infty)$ satisfying:

1. $d(x, y) = 0$ iff $x = y$;
2. $d(x, y) = d(y, x)$ (symmetry);
3. $d(x, z) \leq d(x, y) + d(y, z)$ (triangle inequality).

The open ball of radius $r > 0$ around x is $B_r(x) = \{y : d(x, y) < r\}$. A set is open if it is a union of open balls. A topological space is a set equipped with a collection of open sets satisfying the standard axioms (see Chapter 1 for details).

A sequence (x_n) converges to x if $d(x_n, x) \rightarrow 0$. A Cauchy sequence satisfies $d(x_n, x_m) \rightarrow 0$ as $n, m \rightarrow \infty$. A metric space is *complete* if every Cauchy sequence converges. A set is *compact* if every open cover has a finite subcover; in metric spaces, compactness is equivalent to sequential compactness (every sequence has a convergent subsequence).

In finite-dimensional spaces, compact sets are exactly those that are closed and bounded (Heine-Borel). In infinite-dimensional spaces, this fails: the closed unit ball is not compact. This fact is the source of many of the subtleties in functional analysis.

Later we shall use: completeness ensures that limits exist; compactness (or its failure) determines whether optimization problems attain their extrema. The failure of the Heine-Borel theorem in infinite dimensions is why weak topologies become essential later in this chapter.

5.2 Normed Spaces

Normed spaces are vector spaces equipped with a notion of length. They generalize Euclidean space to infinite dimensions.

Definition 5.2.1. A *normed space* is a vector space X over $\mathbb{C}$ (or $\mathbb{R}$) with a function $\|\cdot\| : X \rightarrow [0, \infty)$ such that:

1. $\|x\| = 0$ iff $x = 0$;
2. $\|\alpha x\| = |\alpha| \|x\|$;
3. $\|x + y\| \leq \|x\| + \|y\|$ (triangle inequality).

The metric is $d(x, y) = \|x - y\|$.

The inverse triangle inequality states $|\|x\| - \|y\|| \leq \|x - y\|$. The norm, vector addition, and scalar multiplication are continuous operations.

In finite-dimensional spaces, all norms are equivalent: for any two norms $\|\cdot\|_1$ and $\|\cdot\|_2$, there exist constants $c_1, c_2 > 0$ such that $c_1\|x\|_1 \leq \|x\|_2 \leq c_2\|x\|_1$. This fails in infinite dimensions, where different norms define genuinely different topologies.

A norm is associated with an inner product iff it satisfies the parallelogram law:

$$\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2. \quad (5.1)$$

Later we shall use: norms quantify distances between quantum states and operators; norm convergence is the strongest form of convergence for sequences of operators. The failure of norm equivalence in infinite dimensions is why we need multiple notions of convergence.

5.3 Banach Spaces

Banach spaces are complete normed spaces. Most infinite-dimensional spaces of interest in functional analysis are Banach spaces.

Definition 5.3.1. A *Banach space* is a normed space that is complete as a metric space.

Standard examples include:

- $C(I)$: continuous functions on a compact interval I with the sup norm $\|f\|_\infty = \max_{x \in I} |f(x)|$. By the Weierstrass theorem, polynomials are dense in $C(I)$; hence $C(I)$ is separable.
- $\ell^1(\mathbb{N}) = \{(a_j) : \sum_j |a_j| < \infty\}$ with $\|a\|_1 = \sum_j |a_j|$.
- $\ell^\infty(\mathbb{N}) = \{(a_j) : \sup_j |a_j| < \infty\}$ with $\|a\|_\infty = \sup_j |a_j|$. This space is not separable.

A series $\sum_j x_j$ in a Banach space is *absolutely convergent* if $\sum_j \|x_j\| < \infty$. Every absolutely convergent series converges.

The Baire category theorem states that in a complete metric space, the intersection of countably many dense open sets is dense. This fundamental result underlies the uniform boundedness principle and the open mapping theorem.

A Banach space is *separable* if it contains a countable dense subset. Many of the spaces encountered in quantum information are separable.

Later we shall use: Banach spaces are the natural setting for most functional analytic arguments; the Baire category theorem (stated without proof) underpins many of the structural results used in operator theory. Separability ensures that sequences suffice for many arguments.

5.4 Hilbert Spaces

Hilbert spaces are the infinite-dimensional analogue of Euclidean spaces. They are the primary setting for quantum mechanics in infinite dimensions.

Definition 5.4.1. A *Hilbert space* is a complete inner product space. The inner product $\langle \cdot, \cdot \rangle$ induces the norm $\|x\| = \sqrt{\langle x, x \rangle}$.

The Cauchy-Schwarz inequality holds:

$$|\langle x, y \rangle| \leq \|x\| \|y\|. \quad (5.2)$$

The polarization identity recovers the inner product from the norm:

$$\langle x, y \rangle = \frac{1}{4} (\|x + y\|^2 - \|x - y\|^2 + i\|x - iy\|^2 - i\|x + iy\|^2). \quad (5.3)$$

Standard examples:

- $\ell^2(\mathbb{N}) = \{(a_j) : \sum_j |a_j|^2 < \infty\}$ with $\langle a, b \rangle = \sum_j a_j^* b_j$. This is a separable Hilbert space.
- $L^2(X, d\mu)$: square-integrable functions on a measure space, with inner product $\langle f, g \rangle = \int_X f(x)^* g(x) d\mu(x)$.

In a Hilbert space, for any closed subspace M , every vector x has a unique decomposition $x = y + z$ with $y \in M$ and $z \perp M$. This is the projection theorem. The map $x \mapsto y$ is the orthogonal projection onto M .

Later we shall use: Hilbert spaces describe the state space of quantum systems in infinite dimensions; the projection theorem underlies the structure of quantum measurements and the decomposition of operators. The self-duality of Hilbert spaces (via the Riesz theorem) is their most important property.

5.5 Completion

Every normed space can be completed to a Banach space, and every inner product space can be completed to a Hilbert space.

Theorem 5.5.1. Every normed space X has a completion $\overline{X}$: a Banach space containing X as a dense subspace. If X is an inner product space, $\overline{X}$ is a Hilbert space. The completion is unique up to isometric isomorphism.

The construction is standard: take equivalence classes of Cauchy sequences, with two sequences equivalent if their difference converges to zero. The norm extends by continuity.

This is exactly how L^2 spaces are obtained from continuous functions: $L^2(I)$ is the completion of $C(I)$ with respect to the inner product $\langle f, g \rangle = \int_I f(x)^* g(x) dx$.

Later we shall use: completion allows us to work with dense subspaces of smooth functions and then extend results by continuity to all of L^2 , a technique used throughout analysis and quantum physics.

5.6 Bounded Operators

Finite-dimensional matrices become bounded linear operators on Banach or Hilbert spaces. Bounded operators are precisely those that are continuous.

Definition 5.6.1. Let X, Y be normed spaces. A linear operator $A : X \rightarrow Y$ is *bounded* if

$$\|A\| = \sup_{\|x\|=1} \|Ax\| < \infty. \quad (5.4)$$

Theorem 5.6.2. A linear operator is bounded iff it is continuous.

The space of bounded operators from X to Y is denoted $\mathcal{L}(X, Y)$. If Y is a Banach space, $\mathcal{L}(X, Y)$ is a Banach space with the operator norm. The space $\mathcal{L}(X)$ of bounded operators on X is a Banach algebra: multiplication (composition) satisfies $\|AB\| \leq \|A\| \|B\|$.

An operator $A \in \mathcal{L}(X)$ is *invertible* if there exists $B \in \mathcal{L}(X)$ with $AB = BA = I$. The Neumann series states that if $\|A\| < 1$, then $I - A$ is invertible with

$$(I - A)^{-1} = \sum_{n=0}^{\infty} A^n, \quad (5.5)$$

and $\|(I - A)^{-1}\| \leq 1/(1 - \|A\|)$.

The B.L.T. theorem states that a bounded operator defined on a dense subspace has a unique extension to the whole space with the same norm.

Later we shall use: bounded operators generalize finite-dimensional matrices to infinite dimensions; quantum channels, when defined on infinite-dimensional Hilbert spaces, are bounded operators; the Neumann series is essential for perturbation theory and the construction of resolvents in Chapter 7.

5.7 Dual Spaces

Dual spaces are the spaces of continuous linear functionals. They are essential for defining weak topologies and for the Riesz representation theorem.

Definition 5.7.1. Let X be a normed space. Its *dual space* is

$$X^* = \mathcal{L}(X, \mathbb{C}), \quad (5.6)$$

the space of all bounded linear functionals on X , equipped with the operator norm:

$$\|\ell\|_{X^*} = \sup_{\|x\|=1} |\ell(x)|. \quad (5.7)$$

The dual of a Banach space is always a Banach space. The second dual X^{**} is the dual of X^* . There is a natural embedding $X \hookrightarrow X^{**}$ given by $x \mapsto \hat{x}$, where $\hat{x}(\ell) = \ell(x)$.

A Banach space is *reflexive* if this embedding is onto (i.e., $X \cong X^{**}$). Hilbert spaces and L^p spaces for $1 < p < \infty$ are reflexive; $C(I)$, ℓ^1 , and L^1 are not.

The dual of ℓ^p for $1 \leq p < \infty$ is ℓ^q where $1/p + 1/q = 1$ (with $q = \infty$ for $p = 1$). The dual of $C(I)$ is the space of regular Borel measures on I (Riesz-Markov theorem).

Later we shall use: dual spaces appear in the formulation of quantum states as positive linear functionals (the GNS construction); the duality between L^1 and L^∞ underlies the distinction between states and observables in infinite-dimensional quantum mechanics.

5.8 Hahn-Banach Theorem

The Hahn-Banach theorem is one of the cornerstones of functional analysis. It guarantees the existence of sufficiently many continuous linear functionals.

Theorem 5.8.1 (Hahn-Banach). Let X be a normed space and let $Y \subseteq X$ be a subspace. Every bounded linear functional $\ell \in Y^*$ has an extension $\tilde{\ell} \in X^*$ with $\|\tilde{\ell}\| = \|\ell\|$.

Corollary 5.8.2. For every nonzero $x \in X$, there exists $\ell \in X^*$ with $\|\ell\| = 1$ and $\ell(x) = \|x\|$.

Corollary 5.8.3. If $\ell(x) = 0$ for all $\ell \in X^*$, then $x = 0$.

The Hahn-Banach theorem is the reason dual spaces are rich enough to separate points. It underlies the separation theorems for convex sets in Banach spaces.

Corollary 5.8.4 (Separation of convex sets). If $C \subset X$ is a closed convex set and $x \notin C$, then there exists $\ell \in X^*$ such that

$$\operatorname{Re} \ell(x) < \inf_{y \in C} \operatorname{Re} \ell(y). \quad (5.8)$$

Later we shall use: Hahn-Banach separates convex sets and is fundamental in duality theory for convex optimization; it also underpins the notion of quantum states as positive linear functionals and the separation of quantum states in resource theories.

5.9 Riesz Representation Theorem

The Riesz representation theorem characterizes the dual of a Hilbert space. This is the single most important theorem connecting Hilbert spaces to their duals.

Theorem 5.9.1 (Riesz representation theorem). Let $\mathcal{H}$ be a Hilbert space. For every bounded linear functional $\ell \in \mathcal{H}^*$, there exists a unique vector $y \in \mathcal{H}$ such that

$$\ell(x) = \langle y, x \rangle \quad \forall x \in \mathcal{H}. \quad (5.9)$$

Moreover, $\|\ell\| = \|y\|$. Thus $\mathcal{H}^* \cong \mathcal{H}$ via the conjugate-linear isomorphism $y \mapsto \ell_y$, where $\ell_y(x) = \langle y, x \rangle$.

This theorem is why Hilbert spaces are so special: they are self-dual. It also implies that every bounded sesquilinear form on a Hilbert space is represented by a bounded operator.

Corollary 5.9.2. For any bounded sesquilinear form $B : \mathcal{H} \times \mathcal{H} \rightarrow \mathbb{C}$, there exists a unique bounded operator A such that $B(x, y) = \langle x, Ay \rangle$.

The Riesz theorem is the infinite-dimensional analogue of the fact that every linear functional on $\mathbb{C}^n$ is given by the inner product with a fixed vector. This is why quantum states in infinite dimensions can still be represented as vectors in a Hilbert space.

Later we shall use: the Riesz theorem identifies states in infinite-dimensional quantum mechanics with density operators (via the trace duality); it underlies the representation of quantum channels as completely positive maps and the structure of the trace duality.

5.10 Weak and Weak-* Topologies

Weak convergence is the appropriate notion of convergence in infinite-dimensional spaces when the usual norm convergence is too strong. It is essential for compactness arguments.

Definition 5.10.1. A sequence $(x_n) \subset X$ converges weakly to $x \in X$, denoted $x_n \rightharpoonup x$, if

$$\ell(x_n) \rightarrow \ell(x) \quad \forall \ell \in X^*. \quad (5.10)$$

The *weak topology* is the weakest topology making all $\ell \in X^*$ continuous.

Similarly, a sequence $(\ell_n) \subset X^*$ converges *weak-** to $\ell \in X^*$ if

$$\ell_n(x) \rightarrow \ell(x) \quad \forall x \in X. \quad (5.11)$$

Key facts:

- Norm convergence implies weak convergence, but the converse is false in infinite dimensions.
- In reflexive spaces, weak and weak-* convergence coincide (for sequences in the space).
- The closed unit ball in a reflexive space is weakly compact.
- The closed unit ball in the dual of a separable space is weak-* compact (Banach-Alaoglu theorem).

These compactness results are the infinite-dimensional substitutes for the Heine-Borel theorem. They are essential for proving the existence of minimizers in optimization problems.

Weak convergence on a Hilbert space has a concrete characterization: $x_n \rightharpoonup x$ iff $\langle y, x_n \rangle \rightarrow \langle y, x \rangle$ for all $y \in \mathcal{H}$. Moreover, if $x_n \rightharpoonup x$ and $\|x_n\| \rightarrow \|x\|$, then $x_n \rightarrow x$ in norm.

Later we shall use: weak convergence appears in the analysis of sequences of quantum states and channels; the compactness of the state space in the weak-* topology ensures the existence of limits in optimization problems; weak-* compactness underlies the mathematical structure of quantum information theory in infinite dimensions.

5.11 L^p Spaces

L^p spaces are the primary examples of Banach spaces in quantum information, arising as the state spaces for continuous variable systems.

Definition 5.11.1. Let (X, Σ, μ) be a σ -finite measure space. For $1 \leq p < \infty$,

$$L^p(X, d\mu) = \left\{ f : X \rightarrow \mathbb{C} \text{ measurable} : \int_X |f|^p d\mu < \infty \right\} / \sim, \quad (5.12)$$

where $f \sim g$ if $f = g$ almost everywhere. The norm is

$$\|f\|_p = \left(\int_X |f|^p d\mu \right)^{1/p}. \quad (5.13)$$

For $p = \infty$,

$$\|f\|_\infty = \text{ess sup}_x |f(x)| = \inf \{C : |f(x)| \leq C \text{ a.e.}\}. \quad (5.14)$$

Theorem 5.11.2 (Hölder's inequality). For $1 \leq p \leq \infty$ with $1/p + 1/q = 1$,

$$\|fg\|_1 \leq \|f\|_p \|g\|_q. \quad (5.15)$$

Theorem 5.11.3 (Minkowski's inequality). $\|f + g\|_p \leq \|f\|_p + \|g\|_p$.

Thus L^p is a normed space. It is complete, hence a Banach space. For $p = 2$, L^2 is a Hilbert space with inner product

$$\langle f, g \rangle = \int_X f(x)^* g(x) d\mu(x). \quad (5.16)$$

The dual of L^p is L^q for $1 \leq p < \infty$ (with $q = \infty$ for $p = 1$). The space L^2 is self-dual, consistent with the Riesz theorem.

For X a locally compact metric space with a regular Borel measure, continuous functions with compact support $C_c(X)$ are dense in $L^p(X, d\mu)$ for $1 \leq p < \infty$. Smooth functions are dense in L^p as well (by mollification).

Later we shall use: L^2 spaces are the state spaces for continuous variable quantum systems; the duality between L^p spaces appears in the theory of quantum states and channels;

the density of continuous functions allows approximation arguments. The L^p framework will be extended in Chapter 7 to Schatten p -classes on operator spaces.

5.12 Summary

Functional analysis extends finite-dimensional linear algebra to infinite-dimensional spaces, providing the mathematical framework for continuous variable quantum systems and the structural theory of quantum channels.

Metric and topological spaces provide the language of convergence. Normed and Banach spaces generalize Euclidean spaces to infinite dimensions. Hilbert spaces are the natural setting for quantum mechanics, with the Riesz theorem identifying states with vectors. Completion allows us to work with dense subspaces. Bounded operators generalize matrices. Dual spaces, the Hahn-Banach theorem, and weak topologies provide the tools for duality and compactness arguments. L^p spaces provide concrete examples of all these structures.

The tools developed in this chapter—particularly Hilbert spaces, bounded operators, dual spaces, and weak topologies—will be essential in the next chapter, where we study operator theory on Hilbert spaces: adjoints, spectra, compact operators, trace-class operators, and functional calculus. Together, Chapters 2, 6, and 7 form the complete linear-algebraic and functional-analytic foundation of quantum information theory.

Chapter 6

Operator Theory on Hilbert Spaces

We now study bounded operators on Hilbert spaces in depth. They play the role of matrices in infinite-dimensional spaces and form the mathematical language of quantum mechanics and quantum information. This chapter develops the core operator-theoretic tools: adjoints, projections, spectra, compact operators, spectral theorems, functional calculus, positive operators, Schatten classes, and density operators. The concepts introduced here are the infinite-dimensional analogues of the finite-dimensional operator theory from Chapter 2.

The reader is assumed to be familiar with the Hilbert space theory reviewed in Chapter 5. Throughout this chapter, $\mathcal{H}$ denotes a complex Hilbert space (not necessarily finite-dimensional), and $\mathcal{L}(\mathcal{H})$ denotes the space of bounded linear operators on $\mathcal{H}$.

6.1 Operators on Hilbert Spaces

Bounded operators on Hilbert spaces are the infinite-dimensional analogues of matrices. They describe quantum observables, evolutions, and measurements.

Definition 6.1.1. A linear operator $A : \mathcal{H} \rightarrow \mathcal{H}$ is *bounded* if

$$\|A\| = \sup_{\|x\|=1} \|Ax\| < \infty. \quad (6.1)$$

The space of all bounded operators is denoted $\mathcal{L}(\mathcal{H})$.

The space $\mathcal{L}(\mathcal{H})$ is a Banach space with the operator norm. It is also a Banach algebra: for $A, B \in \mathcal{L}(\mathcal{H})$, the product AB (composition) satisfies

$$\|AB\| \leq \|A\| \|B\|, \quad (6.2)$$

and the identity operator I satisfies $\|I\| = 1$.

An operator $A \in \mathcal{L}(\mathcal{H})$ is *invertible* if there exists $B \in \mathcal{L}(\mathcal{H})$ with $AB = BA = I$. The Neumann series states that if $\|A\| < 1$, then $I - A$ is invertible with

$$(I - A)^{-1} = \sum_{n=0}^{\infty} A^n, \quad (6.3)$$

and $\|(I - A)^{-1}\| \leq 1/(1 - \|A\|)$.

Unlike matrices, bounded operators on infinite-dimensional spaces have many subtle properties. Not every operator is bounded; unbounded operators (such as differential operators) require careful domain considerations.

Later we shall use: $\mathcal{L}(\mathcal{H})$ is the natural setting for quantum channels, observables, and symmetries. The operator norm controls the stability of quantum operations. The Neumann series is essential for perturbation theory and the construction of resolvents.

6.2 Adjoint Operators

The adjoint generalizes the Hermitian conjugate of a matrix. It is the fundamental operation that distinguishes Hilbert space operators from general Banach space operators.

Theorem 6.2.1. For every $A \in \mathcal{L}(\mathcal{H})$, there exists a unique operator $A^* \in \mathcal{L}(\mathcal{H})$ such that

$$\langle Ax, y \rangle = \langle x, A^*y \rangle \quad \forall x, y \in \mathcal{H}. \quad (6.4)$$

This operator is the *adjoint* of A .

Proof. For fixed y , the map $x \mapsto \langle Ax, y \rangle$ is a bounded linear functional. By the Riesz representation theorem (Chapter 5), there exists a unique vector A^*y such that $\langle Ax, y \rangle = \langle x, A^*y \rangle$. Linearity and boundedness of A^* follow from the uniqueness and the bounds on A . $\square$

The adjoint satisfies:

$$(A + B)^* = A^* + B^*, \quad (6.5)$$

$$(\alpha A)^* = \bar{\alpha} A^*, \quad (6.6)$$

$$(AB)^* = B^* A^*, \quad (6.7)$$

$$(A^*)^* = A, \quad (6.8)$$

$$\|A^*\| = \|A\|. \quad (6.9)$$

Important classes of operators defined via the adjoint:

Class	Condition
Self-adjoint	$A = A^*$
Unitary	$A^*A = AA^* = I$
Normal	$AA^* = A^*A$
Isometry	$A^*A = I$
Partial isometry	A^*A is a projection

Self-adjoint operators have real spectrum and are the mathematical model for quantum observables. Unitary operators preserve the inner product and describe reversible quantum evolutions.

Later we shall use: adjoints define Hermitian conjugates of quantum channels; self-adjoint operators model measurements; unitaries describe closed system evolution.

6.3 Orthogonal Projections

Projections are the infinite-dimensional generalization of projections onto subspaces. They are fundamental to measurement theory and the decomposition of Hilbert spaces.

Definition 6.3.1. An operator $P \in \mathcal{L}(\mathcal{H})$ is an *orthogonal projection* if

$$P^2 = P \quad \text{and} \quad P^* = P. \quad (6.10)$$

Theorem 6.3.2 (Projection theorem). For every closed subspace $M \subseteq \mathcal{H}$, there exists a unique orthogonal projection P_M onto M . Moreover,

$$\mathcal{H} = M \oplus M^\perp, \quad (6.11)$$

where $M^\perp = \{x : \langle x, y \rangle = 0 \ \forall y \in M\}$.

The projection theorem is the infinite-dimensional analogue of the fact that every subspace of a finite-dimensional inner product space has an orthogonal complement. The closedness condition is essential in infinite dimensions.

Every orthogonal projection P has the properties:

- $0 \leq P \leq I$ (in the sense $\langle x, Px \rangle \leq \langle x, x \rangle$).
- $\|P\| = 1$ (unless $P = 0$).
- $\text{im } P = \ker(I - P) = M$, $\ker P = M^\perp$.

Later we shall use: projections describe quantum measurements; the projection theorem underlies the decomposition of states into pure components.

6.4 Orthonormal Bases

Orthonormal bases provide the coordinate systems for Hilbert spaces, generalizing the standard basis of $\mathbb{C}^n$.

An *orthonormal basis* of $\mathcal{H}$ is a maximal orthonormal set. Every Hilbert space has an orthonormal basis (by Zorn's lemma). For a separable Hilbert space, the basis is countable.

For an orthonormal basis $\{e_n\}_{n \in I}$, every $x \in \mathcal{H}$ has the expansion

$$x = \sum_{n \in I} \langle e_n, x \rangle e_n. \quad (6.12)$$

The Gram-Schmidt process works in any Hilbert space. Parseval's identity holds:

$$\|x\|^2 = \sum_{n \in I} |\langle e_n, x \rangle|^2. \quad (6.13)$$

The Riesz-Fischer theorem states that for any square-summable family of coefficients $\{\alpha_n\}$, there exists a unique $x \in \mathcal{H}$ with $\langle e_n, x \rangle = \alpha_n$.

A Hilbert space is separable iff it has a countable orthonormal basis. Most Hilbert spaces encountered in quantum information are separable.

Later we shall use: orthonormal bases provide the expansion of states in infinite dimensions; Parseval's identity is used to compute norms and traces.

6.5 Compact Operators

Compact operators are the infinite-dimensional analogues of finite-rank operators. They are exactly the operators for which the finite-dimensional spectral theorem extends most naturally.

Definition 6.5.1. An operator $A \in \mathcal{L}(\mathcal{H})$ is *compact* if the image of the closed unit ball is precompact (i.e., its closure is compact). Equivalently, for every bounded sequence (x_n) , (Ax_n) has a convergent subsequence.

Every finite-rank operator is compact. The set of compact operators $\mathcal{K}(\mathcal{H})$ is the closure of the finite-rank operators in the operator norm. It is a closed two-sided ideal in $\mathcal{L}(\mathcal{H})$.

For a compact operator A , the singular values $s_n(A)$ are defined as the eigenvalues of $\sqrt{A^*A}$, which exists by the spectral theorem for positive operators (to be proved in the next section). They form a sequence tending to 0. The operator can be written as

$$A = \sum_{n=1}^{\infty} s_n(A) \langle \psi_n, \cdot \rangle \phi_n, \quad (6.14)$$

where $\{\psi_n\}$ and $\{\phi_n\}$ are orthonormal sets. This is the singular value decomposition of A .

Compact self-adjoint operators have the spectral decomposition

$$A = \sum_n \lambda_n \langle \psi_n, \cdot \rangle \psi_n, \quad (6.15)$$

with $\lambda_n \rightarrow 0$. The nonzero spectrum consists of eigenvalues of finite multiplicity accumulating only at 0.

Later we shall use: compact operators generalize finite-rank operators; they are the building blocks for trace-class and Hilbert-Schmidt operators used in quantum information. The singular value decomposition of compact operators is the infinite-dimensional analogue of the matrix SVD.

6.6 Spectrum and Resolvent

The spectrum generalizes the set of eigenvalues from finite dimensions. In infinite dimensions, the spectrum contains more information than just the eigenvalues.

Definition 6.6.1. For $A \in \mathcal{L}(\mathcal{H})$, the *resolvent set* is

$$\rho(A) = \{\lambda \in \mathbb{C} : \lambda I - A \text{ is invertible with bounded inverse}\}. \quad (6.16)$$

The *spectrum* is $\sigma(A) = \mathbb{C} \setminus \rho(A)$.

The spectrum is always a nonempty compact subset of $\mathbb{C}$ bounded by $\|A\|$. The *spectral radius* is

$$r(A) = \sup\{|\lambda| : \lambda \in \sigma(A)\}. \quad (6.17)$$

One has $r(A) \leq \|A\|$, with equality for normal operators.

The spectrum is partitioned into:

- *Point spectrum:* eigenvalues λ for which $\ker(\lambda I - A) \neq \{0\}$.
- *Continuous spectrum:* $\lambda I - A$ is injective with dense range, but not surjective.
- *Residual spectrum:* $\lambda I - A$ is injective, but the range is not dense.

For self-adjoint operators, the spectrum is real and there is no residual spectrum.

The *resolvent* is $R(z, A) = (zI - A)^{-1}$ for $z \in \rho(A)$. It satisfies the resolvent identity:

$$R(z, A) - R(w, A) = (w - z)R(z, A)R(w, A). \quad (6.18)$$

The resolvent is an analytic function of z on $\rho(A)$.

Examples of spectra:

- Identity operator: $\sigma(I) = \{1\}$.
- Shift operator on ℓ^2 : $\sigma(S) = \overline{\mathbb{D}}$ (the closed unit disk).
- Multiplication operator on L^2 : $\sigma(M_f) = \text{ess ran}(f)$.

Later we shall use: the spectrum determines the possible measurement outcomes for an observable; the spectral radius appears in convergence criteria for operator sequences; the resolvent is the key tool in functional calculus and perturbation theory.

6.7 Spectral Theorem

The spectral theorem is the culmination of operator theory. It generalizes the diagonalization of Hermitian matrices to infinite dimensions.

Theorem 6.7.1 (Spectral theorem for compact self-adjoint operators). Let $A \in \mathcal{K}(\mathcal{H})$ be self-adjoint. Then $\mathcal{H}$ has an orthonormal basis consisting of eigenvectors of A , and

$$A = \sum_{n=1}^{\infty} \lambda_n \langle \psi_n, \cdot \rangle \psi_n, \quad (6.19)$$

where $\lambda_n \in \mathbb{R}$, $\lambda_n \rightarrow 0$, and $\{\psi_n\}$ is an orthonormal basis of $\overline{\text{im}(A)}$.

This is the infinite-dimensional analogue of the finite-dimensional spectral theorem from Chapter 2. The eigenvalues λ_n accumulate only at 0, and 0 may be in the spectrum without being an eigenvalue (if the operator is not surjective).

For general (not necessarily compact) self-adjoint operators, the spectral theorem takes the form

$$A = \int_{\sigma(A)} \lambda dE(\lambda), \quad (6.20)$$

where E is a projection-valued measure (PVM). This integral representation is the quantum mechanical version of diagonalization.

The PVM E assigns to each Borel set $B \subseteq \mathbb{R}$ an orthogonal projection $E(B)$ such that:

- $E(\emptyset) = 0$, $E(\mathbb{R}) = I$;
- $E(B_1 \cap B_2) = E(B_1)E(B_2)$;
- $E(B_1 \cup B_2) = E(B_1) + E(B_2)$ for disjoint B_1, B_2 .

The spectral theorem implies that every self-adjoint operator can be written as the difference of two positive operators (the spectral decomposition).

Later we shall use: the spectral theorem underlies the measurement postulate in quantum mechanics; it is essential for the formulation of quantum dynamics and the definition of functions of observables.

6.8 Functional Calculus

The spectral theorem allows us to define functions of self-adjoint operators. The projection-valued measure formulation of the general spectral theorem immediately leads to the functional calculus.

For a self-adjoint operator $A = \int \lambda dE(\lambda)$ and a measurable function $f : \sigma(A) \rightarrow \mathbb{C}$, one defines

$$f(A) = \int_{\sigma(A)} f(\lambda) dE(\lambda). \quad (6.21)$$

For polynomials, this agrees with the algebraic definition. For continuous functions, it gives a continuous map

$$C(\sigma(A)) \rightarrow \mathcal{L}(\mathcal{H}), \quad f \mapsto f(A), \quad (6.22)$$

with $\|f(A)\| \leq \|f\|_\infty$.

Important examples:

- Exponential: $\exp(A) = \int e^\lambda dE(\lambda)$.
- Square root: for $A \geq 0$, $\sqrt{A} = \int \sqrt{\lambda} dE(\lambda)$.
- Logarithm: for $A > 0$, $\log A = \int \log \lambda dE(\lambda)$.
- Sign: $\operatorname{sgn}(A) = \int \operatorname{sgn}(\lambda) dE(\lambda)$.
- Spectral projections: $E(-\infty, a] = \chi_{(-\infty, a]}(A)$.

The functional calculus is the mathematical basis for defining functions of density operators, Hamiltonians, and other quantum observables.

Later we shall use: functional calculus defines $\sqrt{\rho}$ for fidelity, $\log \rho$ for entropy, $\exp(-itH)$ for dynamics, and spectral projections for measurements.

6.9 Positive Operators

Positive operators are the infinite-dimensional generalization of positive semidefinite matrices. They are fundamental to quantum states and measurements.

Definition 6.9.1. An operator $A \in \mathcal{L}(\mathcal{H})$ is *positive* if

$$A = A^* \quad \text{and} \quad \langle x, Ax \rangle \geq 0 \quad \forall x \in \mathcal{H}. \quad (6.23)$$

We write $A \geq 0$.

Equivalently, $A \geq 0$ iff its spectrum is contained in $[0, \infty)$. For self-adjoint operators, this follows from the spectral theorem.

The functional calculus gives the positive square root: for $A \geq 0$, there exists a unique $B \geq 0$ such that $B^2 = A$. We write $B = A^{1/2}$.

For self-adjoint operators A, B , we write $A \leq B$ if $B - A \geq 0$. This defines a partial order on $\text{Herm}(\mathcal{H})$.

The positive cone has the following properties:

- If $A \geq 0$ and $\lambda \geq 0$, then $\lambda A \geq 0$.
- If $A, B \geq 0$, then $A + B \geq 0$.
- If $A \geq 0$ and $T \in \mathcal{L}(\mathcal{H})$, then $T^*AT \geq 0$.
- If $A \geq 0$ and A is invertible, then $A^{-1} \geq 0$.

The polar decomposition states that every $A \in \mathcal{L}(\mathcal{H})$ can be written uniquely as $A = U|A|$, where $|A| = \sqrt{A^*A}$ and U is a partial isometry.

Later we shall use: positive operators form the cone from which density operators are chosen; the square root defines fidelity; the partial order $A \leq B$ appears in operator inequalities throughout quantum information.

6.10 Schatten Classes

The Schatten classes generalize the trace, Hilbert-Schmidt, and operator norms to a unified family. They are essential for quantum states and channels.

For $1 \leq p < \infty$, the *Schatten p -class* is

$$\mathcal{S}_p(\mathcal{H}) = \left\{ A \in \mathcal{K}(\mathcal{H}) : \|A\|_p = \left(\sum_n s_n(A)^p \right)^{1/p} < \infty \right\}, \quad (6.24)$$

where $s_n(A)$ are the singular values. For $p = \infty$, $\mathcal{S}_\infty(\mathcal{H}) = \mathcal{L}(\mathcal{H})$ with $\|A\|_\infty = \|A\|$.

The Schatten classes form a hierarchy:

$$\text{Finite rank} \subset \mathcal{S}_1 \subset \mathcal{S}_2 \subset \cdots \subset \mathcal{S}_p \subset \mathcal{S}_q \subset \mathcal{K} \subset \mathcal{L} \quad (6.25)$$

for $1 \leq p < q < \infty$.

- $\mathcal{S}_1$: *trace-class operators*. For $A \in \mathcal{S}_1$, the trace

$$\text{tr}(A) = \sum_n \langle A e_n, e_n \rangle \quad (6.26)$$

is well-defined and independent of the orthonormal basis. It satisfies the cyclic property:

$$\text{tr}(AB) = \text{tr}(BA) \quad (6.27)$$

for $A \in \mathcal{S}_1$, $B \in \mathcal{L}(\mathcal{H})$.

- $\mathcal{S}_2$: *Hilbert-Schmidt operators*. They form a Hilbert space with inner product $\langle A, B \rangle_2 = \text{tr}(A^*B)$. Hilbert-Schmidt operators play an important role in the study of quantum channels and operator representations.
- $\mathcal{S}_\infty = \mathcal{L}(\mathcal{H})$: bounded operators with the operator norm.

Duality relations:

- $(\mathcal{S}_1)^* \cong \mathcal{L}(\mathcal{H})$ via $\langle A, B \rangle = \text{tr}(AB)$.
- $(\mathcal{S}_2)^* \cong \mathcal{S}_2$ via the Hilbert-Schmidt inner product.
- For $1 < p < \infty$, $(\mathcal{S}_p)^* \cong \mathcal{S}_q$ with $1/p + 1/q = 1$.

Later we shall use: trace-class operators describe quantum states; Hilbert-Schmidt operators appear in the representation of quantum channels; the Schatten hierarchy provides the framework for quantum state discrimination and channel analysis.

6.11 Density Operators

Density operators are the mathematical model for quantum states. In infinite dimensions, they are positive trace-class operators with trace one.

Definition 6.11.1. A *density operator* is an operator $\rho \in \mathcal{S}_1(\mathcal{H})$ such that

$$\rho \geq 0, \quad \text{tr}(\rho) = 1. \quad (6.28)$$

The set of all density operators is denoted $\mathcal{D}(\mathcal{H})$.

It is important to emphasize: $\rho \in \mathcal{S}_1(\mathcal{H})$, not merely $\rho \in \mathcal{L}(\mathcal{H})$. The trace-class condition ensures that $\text{tr}(\rho)$ is well-defined.

Pure states correspond to rank-one orthogonal projections: $\rho = |\psi\rangle\langle\psi|$ with $\|\psi\| = 1$. Mixed states are convex combinations of pure states.

A density operator admits a spectral decomposition:

$$\rho = \sum_n p_n |\psi_n\rangle\langle\psi_n|, \quad (6.29)$$

where $p_n \geq 0$, $\sum_n p_n = 1$, and $\{\psi_n\}$ is an orthonormal basis. This is the infinite-dimensional version of the finite-dimensional density matrix.

An *observable* is a self-adjoint operator $A \in \mathcal{L}(\mathcal{H})$. The expectation value of A in state ρ is

$$\langle A \rangle_\rho = \text{tr}(\rho A). \quad (6.30)$$

Later we shall use: density operators describe quantum states in infinite dimensions. The trace-class property ensures finite expectation values. This framework provides the foundation for quantum information theory.

6.12 Summary

Operator theory extends the finite-dimensional operator algebra of Chapter 2 to infinite-dimensional Hilbert spaces, providing the mathematical foundation for quantum mechanics and quantum information in continuous variable systems.

Bounded operators generalize matrices; the adjoint generalizes Hermitian conjugation. Orthogonal projections decompose Hilbert spaces and describe measurements. Orthonormal bases provide coordinate systems. Compact operators are the natural infinite-dimensional analogue of finite-rank operators. The spectrum generalizes the set of eigenvalues, and

the spectral theorem provides the canonical decomposition of self-adjoint operators via projection-valued measures. The functional calculus allows functions of operators to be defined. Positive operators form the cone from which quantum states are chosen. Schatten classes unify trace-class, Hilbert-Schmidt, and bounded operators into a single hierarchy. Density operators are positive trace-one elements of the trace class, providing the mathematical model for quantum states.

Operator theory is the mathematical bridge between functional analysis and quantum information. While Chapter 5 develops the geometry of infinite-dimensional spaces, this chapter develops the algebra of operators acting on those spaces. Together, Chapters 2, 5, and 6 form the complete mathematical foundation of quantum information theory.